



DEPARTMENT OF COMPUTING
AND CONTROL ENGINEERING

ДЕПАРТАМЕНТ ЗА РАЧУНАРСТВО
И АУТОМАТИКУ

ACSE

Odsek za automatiku, geomatiku
i upravljanje sistemima

Sub- department for Automatic Control
and System Engineering

Modeli jednostavnih električnih i elektro-mehaničkih sistema

Modeli fizičkih sistema

Modeliranje i simulacija sistema

Upravljanje, modelovanje i simulacija sistema

Električni sistemi

Veličine i elementi

Osnovne fizičke veličine:

- napon u [V] i
- jačina električne struje i [A].
- električna snaga $p = ui$ [W] i
- električna energija $E = \int p(t)dt$

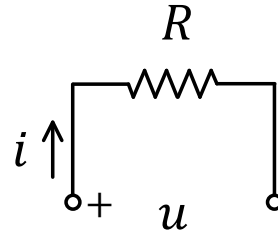
Pasivni elementi:

- otpornik otpornosti R [Ω]
- kalem induktivnosti L [H]
- kondenzator kapaciteta C [F]

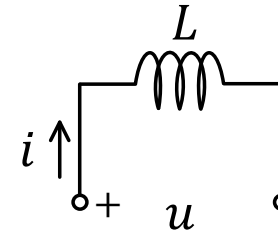
$$u(t) = Ri(t)$$

$$u(t) = L \frac{di_L(t)}{dt}$$

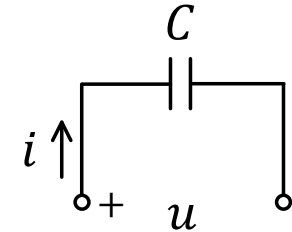
$$i(t) = C \frac{du_C(t)}{dt}$$



Otpornik



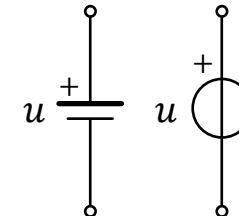
Kalem



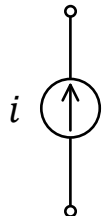
Kondenzator

Aktivni elementi:

- Idealan naponski izvor
- Idealan strujni izvor



Idealni
naponski izvori



Idealan
strujni izvor

Zakovitosti

- Omov zakon

$$i = \frac{u}{R}$$

- I Kirhofov zakon: algebarska suma struja koje utiču u ma koji čvor kola jednaka je nuli

$$\sum_{\text{po granama } k \text{ u čvoru}} i_k = 0$$

- II Kirhofov zakon: algebarska suma padova napona u bilo kojoj konturi (petlji) električnog kola ja jednaka nuli

$$\sum_{\text{po elementima } k \text{ u konturi}} u_k = 0$$

- Alternative tehnike formiranja modela

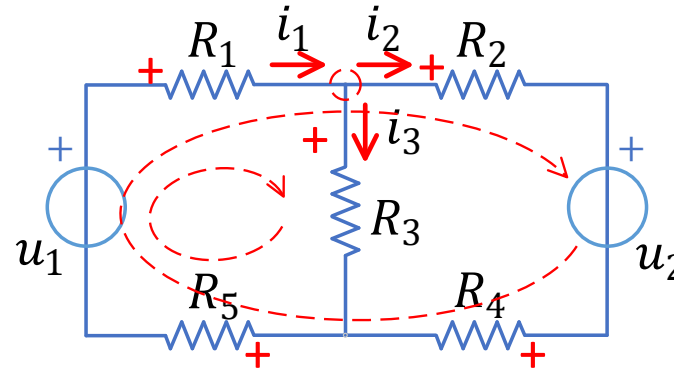
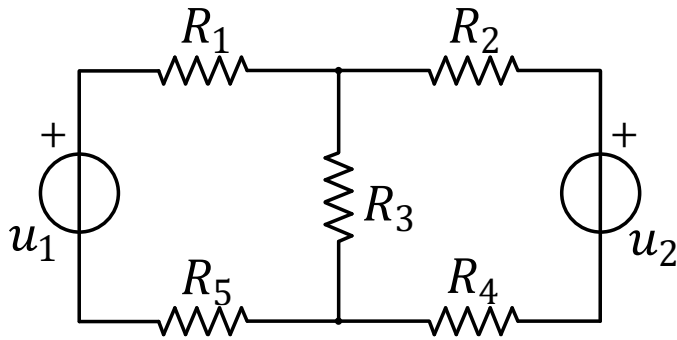
- Metoda potencijala čvorova
- Metoda konturnih struja

upotrebljavaju se često jer se dobija manji broj jednačina u odnosu na direktnu upotrebu Kirhofovih zakona

Primer 1 – Električno kolo sa otpornicima

Zadatak:

Napisati model za el.kolo sa slike.



Formiranje modela primenom Kirhofovih zakona:

$$i_1 - i_2 - i_3 = 0$$

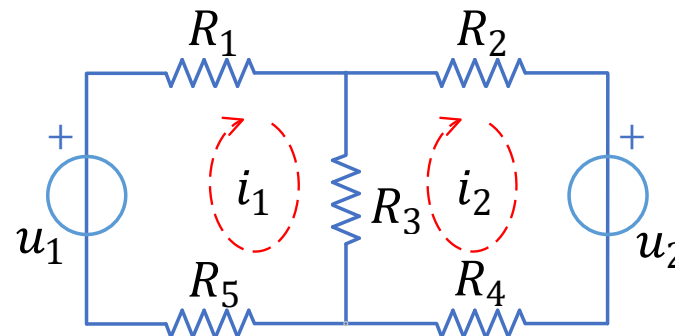
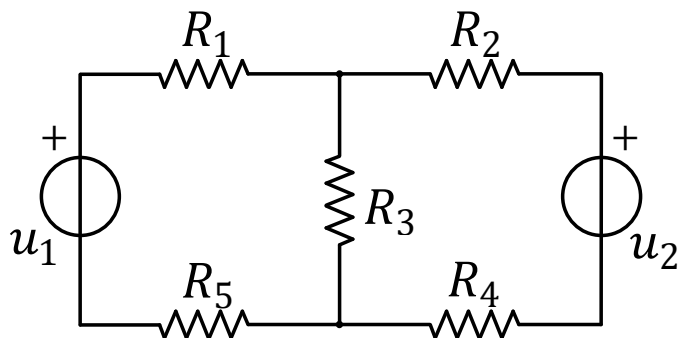
$$u_1 - (R_1 + R_5)i_1 - R_3i_3 = 0$$

$$u_1 - (R_1 + R_5)i_1 - (R_2 + R_4)i_2 - u_2 = 0$$

U matričnom zapisu:

$$\begin{bmatrix} 1 & -1 & -1 \\ -R_1 - R_5 & 0 & -R_3 \\ -R_1 - R_5 & -R_2 - R_4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ u_1 \\ u_1 - u_2 \end{bmatrix}$$

Primer 1 – nastavak ...



Konturne struje:

$$R_1 i_1 + R_3 (i_1 - i_2) + R_5 i_1 = u_1$$

$$R_4 i_2 + R_3 (i_2 - i_1) + R_2 i_2 = -u_2$$

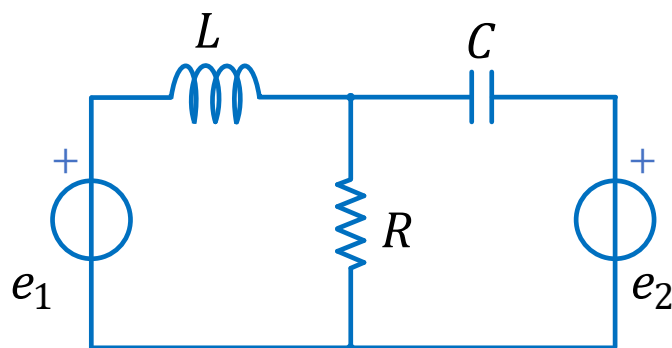
Matrični zapis:

$$\begin{bmatrix} R_1 + R_3 + R_5 & -R_3 \\ -R_3 & R_2 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} u_1 \\ -u_2 \end{bmatrix}$$

Primer 2 - Električno kolo sa R, L i C

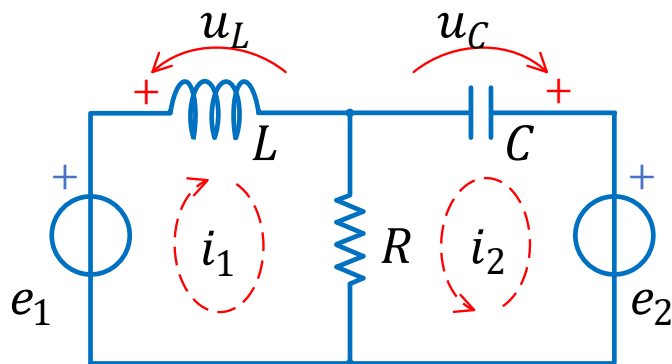
Zadatak:

Napisati model za el.kolo sa slike.



- Formirati matematički model u prostoru stanja za električno kolo na slici.
- Ako se za izlaznu veličinu usvoji struja kroz otpornik R , formirati jednačinu izlaza.
- Formirati jednačinu izlaza ako se za izlazne veličine usvoje struje kroz izvore e_1 i e_2 .

Primer 2 – nastavak ...



$$i_2 = C \frac{du_C}{dt}$$

$$u_L = L \frac{di_1}{dt}$$

- Konturne struje

$$e_2 - R(i_1 + i_2) - u_C = 0 \quad \rightarrow \quad R(i_1 + i_2) = e_2 - u_C$$

$$e_1 - u_L - R(i_1 + i_2) = 0 \quad \rightarrow \quad e_1 - u_L - e_2 + u_C = 0$$

- Uvedemo izvide

$$R \left(i_1 + C \frac{du_C}{dt} \right) = e_2 - u_C$$

$$e_1 - L \frac{di_1}{dt} - e_2 + u_C = 0$$

- prepišemo ...

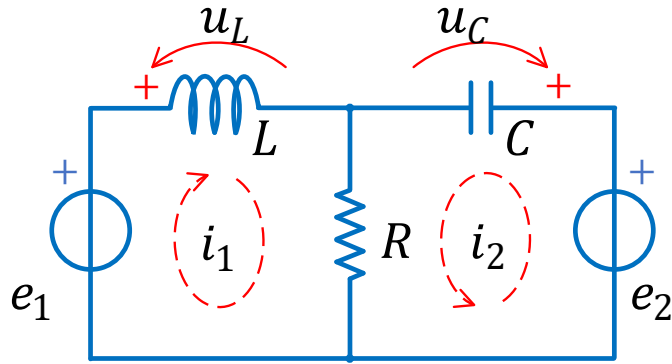
$$\frac{du_C}{dt} = -\frac{1}{RC} u_C - \frac{1}{C} i_1 + \frac{1}{RC} e_2$$

$$\frac{di_1}{dt} = \frac{1}{L} u_C + \frac{1}{L} e_1 - \frac{1}{L} e_2$$

- Matrični zapis

$$\begin{bmatrix} \frac{di_1}{dt} \\ \frac{du_C}{dt} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} & -\frac{1}{L} \\ 0 & \frac{1}{RC} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

Primer 2 – nastavak 2 ...



$$i_2 = C \frac{du_C}{dt}$$
$$u_L = L \frac{di_1}{dt}$$

- Izlaz: struja kroz otpornik

$$i = i_1 + i_2$$

$$R(i_1 + i_2) = e_2 - u_C \rightarrow$$

$$i = \frac{e_2 - u_C}{R}$$

$$i = \begin{bmatrix} 0 & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

- Izlazi: struje kroz naponske izvore

$$i_2 = C \frac{du_C}{dt} = -\frac{1}{R} u_C - i_1 + \frac{1}{R} e_2$$

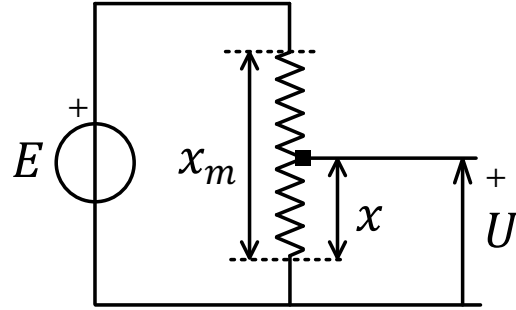
$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{R} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

Elektro-mehanički sistemi

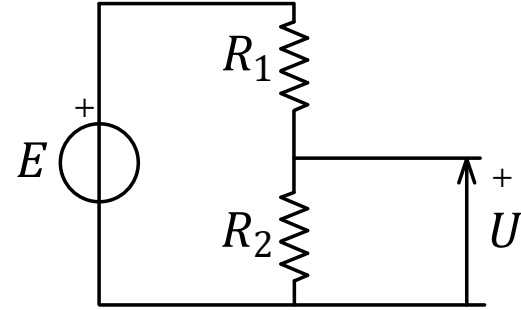
Interakcija električnih i mehaničkih sistema

- Električni i mehanički deo sistema mogu imati različite vidove interakcije:
 - mehanička promena otpornosti,
 - kretanje strujnog provodnika u magnetnom polju.

Potenciometar



$$R(x) = \frac{\rho}{A} x$$
$$R(x) = \frac{R_m}{x_m} x$$



$$U(t) = \frac{R_2}{R_1 + R_2} E(t)$$

Promenljive

Promenljive od značaja:

- f_e – elektromagnetna sila $[N]$
- v – brzina provodnika u odnosu na magnetno polje $[m/s]$
- l – dužina provodnika u magnetnom polju $[m]$
- Φ – magnetni fluks $[Wb]$ (jedinica je Veber)
- B – magnetna indukcija $[T]$ (jedinica je Tesla) $[T] = [Wb/m^2]$
- i – jačina električne struje $[A]$
- e_m – indukovana elektromotorna sila $[V]$

Kretanje provodnika u magnetnom polju

- Na pravolinijski provodnik dužine l , kroz koji protiče struja jačine i i koji se kreće u homogenom magnetnom polju indukcije B deluje sila (Amperova sila) intenziteta

$$\vec{f}_e = i\vec{l} \times \vec{B}$$

θ ugao koji zaklapaju silnice magnetnog polja i smer proticanja struje

$$f_e = ilB \sin \theta$$

$$\text{za } \theta = 90^\circ: f_e = ilB$$

- kretanje provodnika brzinom v u homogenom magnetnom indukuje elektromotornu silu

$$e_m = l\vec{v} \times \vec{B}.$$

Kada se provodnik kreće normalno na silnice magnetnog polja

$$e_m = lvB$$

- Dvosmerno dejstvo:

$$p_m = f_e v = (Bli)v$$

$$p_e = e_m i = (Blv)i$$

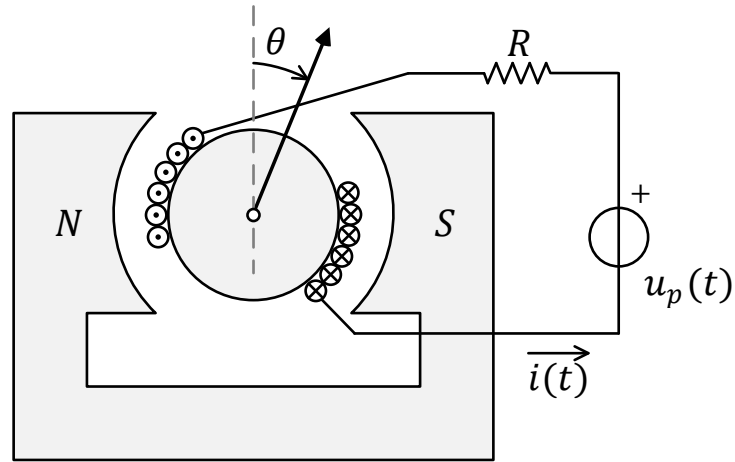
$$p_m = p_e$$

mehanička snaga - magnetno polje deluje na mehaničke elemente

električna snaga - proticanje struje usled e_m

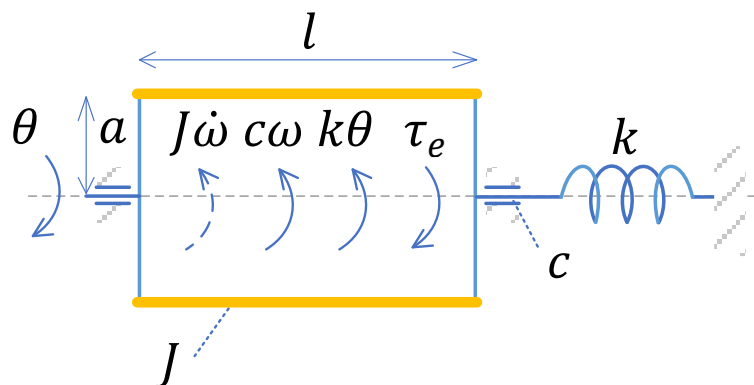
iz zakona održanja energija.

Primer 1 - Galvanometar



- Ugaoni pomeraj kalema je srazmeran struji koja protekne kroz namotaj
- U osi kalema postoji torziona opruga (nije nacrtana)
- Pretpostavka: u vazдушnom procepu je uniforman mag. fluks gustine B
- Ima n namotaja žice (dužine l , poluprečnika a)

Primer 1 – nastavak ...



Mehanički deo

$$f_e = (2nl)Bi$$

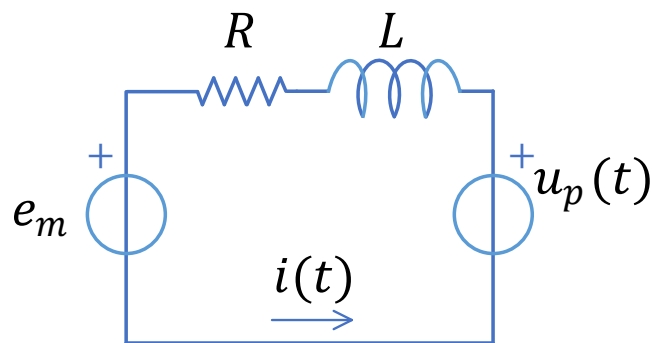
sila na provodnike u namotaju

$$\tau_e = f_e a = (2nl)Bia$$

moment sile

Ravnoteža momenata sile

$$J\dot{\omega}(t) + c\omega(t) + k\theta(t) = (2nlBa)i(t)$$



Električni deo

$$e_m = (2nl)vB = (2nl)(a\omega)B$$

Indukovana ems

El. kolo

$$L \frac{di(t)}{dt} + Ri(t) + (2nlBa)\omega(t) = u_p(t)$$

Primer 1 - nastavak 2

prepisano ...

$$J\dot{\omega}(t) + c\omega(t) + k\theta(t) = (2nlBa)i(t)$$

$$L \frac{di(t)}{dt} + Ri(t) + (2nlBa)\omega(t) = u_p(t)$$

Uvedemo: $\alpha = 2nlBa$

Model (sređeno):

$$\frac{d\theta}{dt}(t) = \omega(t)$$

$$\frac{d\omega}{dt}(t) = -\frac{c}{J}\omega(t) - \frac{k}{J}\theta(t) + \frac{\alpha}{J}i(t)$$

$$\frac{di}{dt}(t) = -\frac{R}{L}i(t) - \frac{\alpha}{L}\omega(t) + \frac{1}{L}u_p(t)$$

Model 3. reda

Ako zanemarimo induktivnost kalema $L \ll 1$

$$\underbrace{L \frac{di(t)}{dt}}_0 + Ri(t) + \alpha\omega(t) = u_p(t)$$

$$i(t) = -\frac{\alpha}{R}\omega(t) + \frac{1}{R}u_p(t)$$

$$\ddot{\theta}(t) + \left(\frac{c}{J} + \frac{\alpha^2}{JR}\right)\dot{\theta}(t) + \frac{k}{J}\theta(t) = \frac{\alpha}{JR}u_p(t)$$

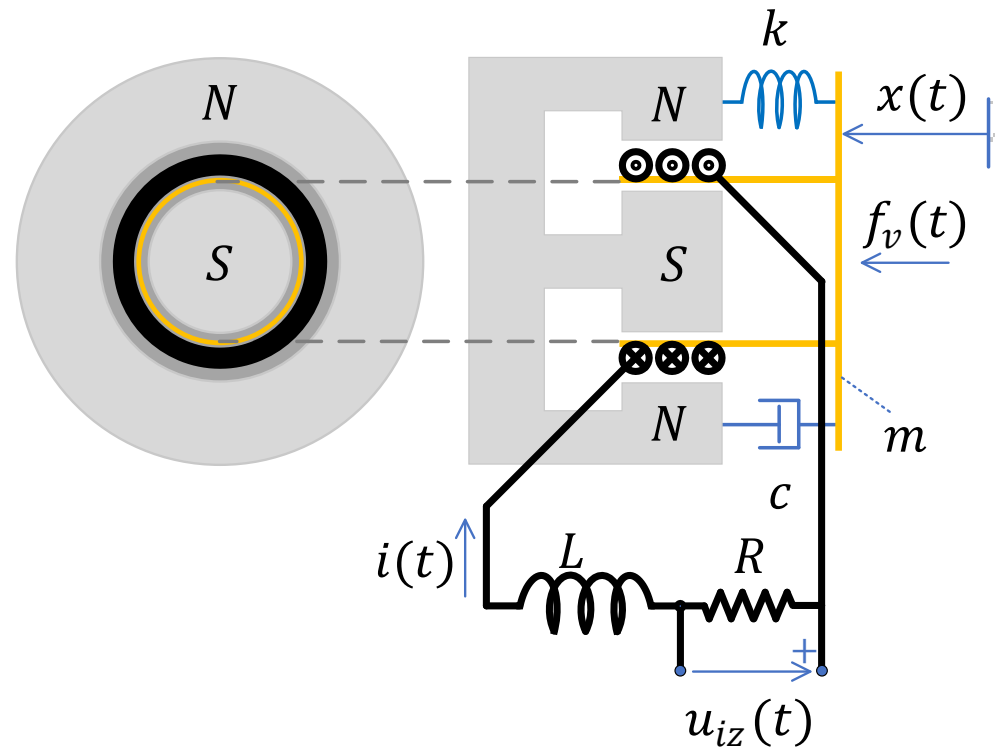
Model 2. reda

Analiza ponašanja:

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

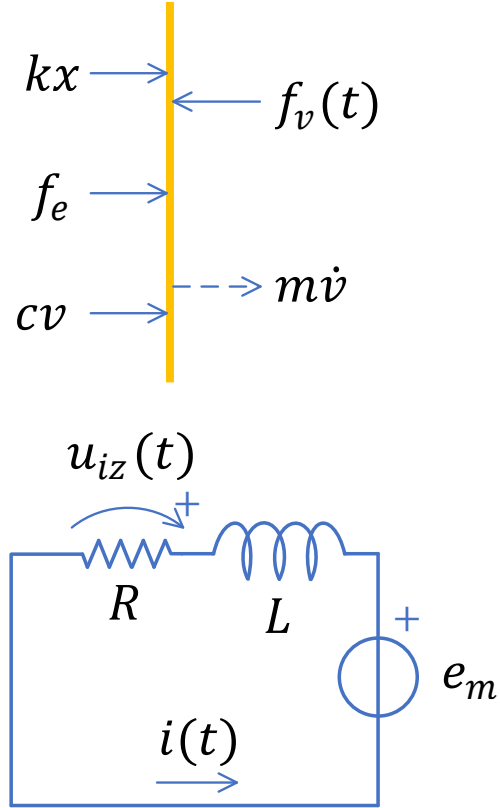
$$\xi = \frac{1}{2\sqrt{kJ}}\left(c + \frac{\alpha^2}{R}\right), \quad \omega_n = \sqrt{\frac{k}{J}}.$$

Primer 2 - Mikrofon



- Ima n namotaja žice u kalemu poluprečnika a
- Otpornost namotaja se zanemaruje
- Radijalno mag. polje na namotaje

Primer 2 – nastavak



$$f_e = lBi(t) = (2\pi an)Bi(t) = \alpha \cdot i(t)$$

$$e_m = lBv = (2\pi an)Bv(t) = \alpha \cdot v(t)$$

$$\alpha = (2\pi an)B$$

Mehanički deo: $m\dot{v}(t) + cv(t) + kx(t) = -\alpha i(t) + f_v(t)$

Električni deo: $L \frac{di}{dt} + Ri(t) = \alpha v(t)$

Model:

$$\frac{dx(t)}{dt} = v(t)$$

$$\frac{dv(t)}{dt} = \frac{1}{m}(-kx(t) - cv(t) - \alpha i(t) + f_v(t))$$

$$\frac{di(t)}{dt} = \frac{1}{L}(\alpha v(t) - Ri(t))$$

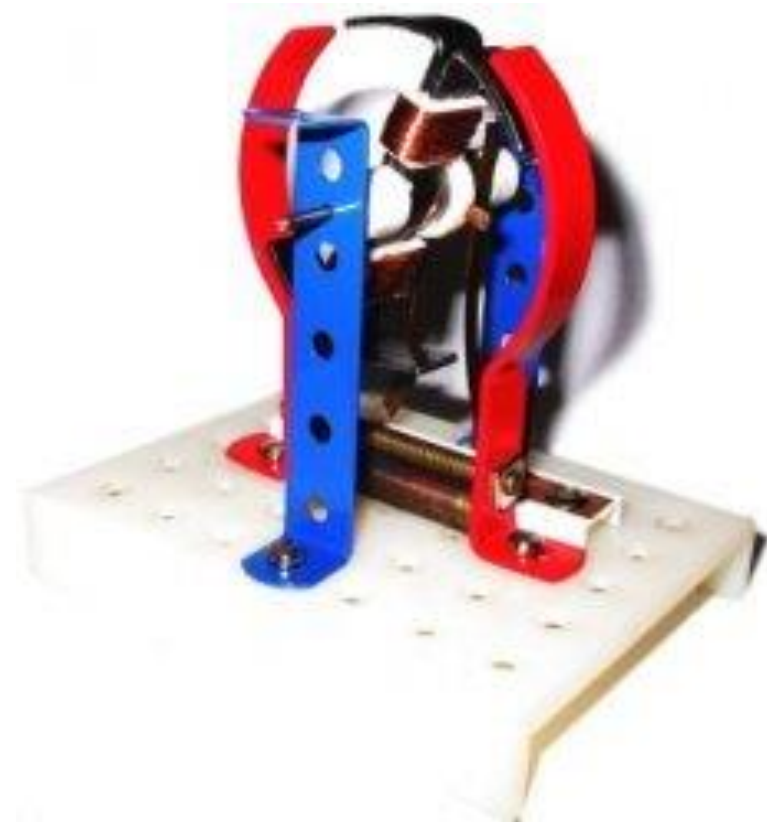
$$u_{iz}(t) = Ri(t)$$

Matrični zapis:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{k}{m} & -\frac{c}{m} & -\frac{\alpha}{m} \\ 0 & \frac{\alpha}{L} & -\frac{R}{L} \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m} \end{bmatrix} f_v(t)$$

$$u_{iz}(t) = [0 \quad 0 \quad R] \mathbf{x}(t) + 0 \cdot f_v(t)$$

Elektromotor



DC Machine Construction

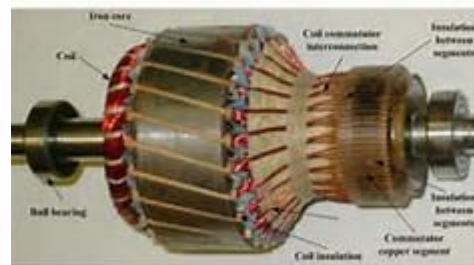
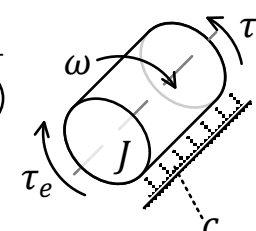
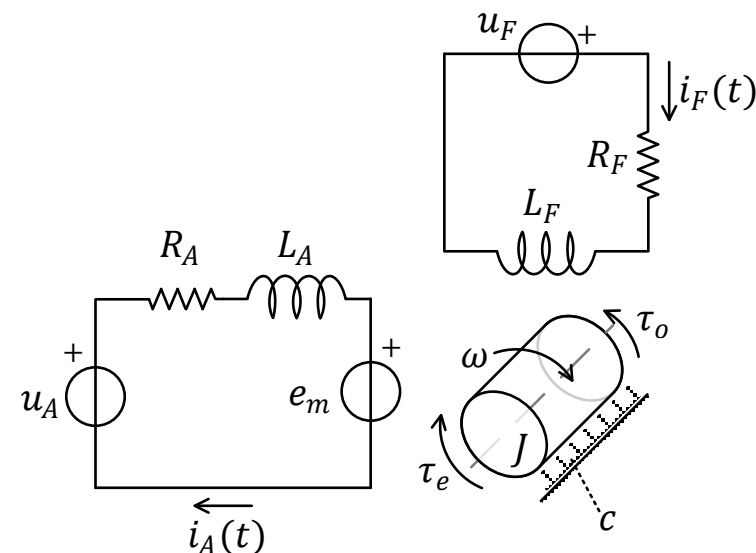
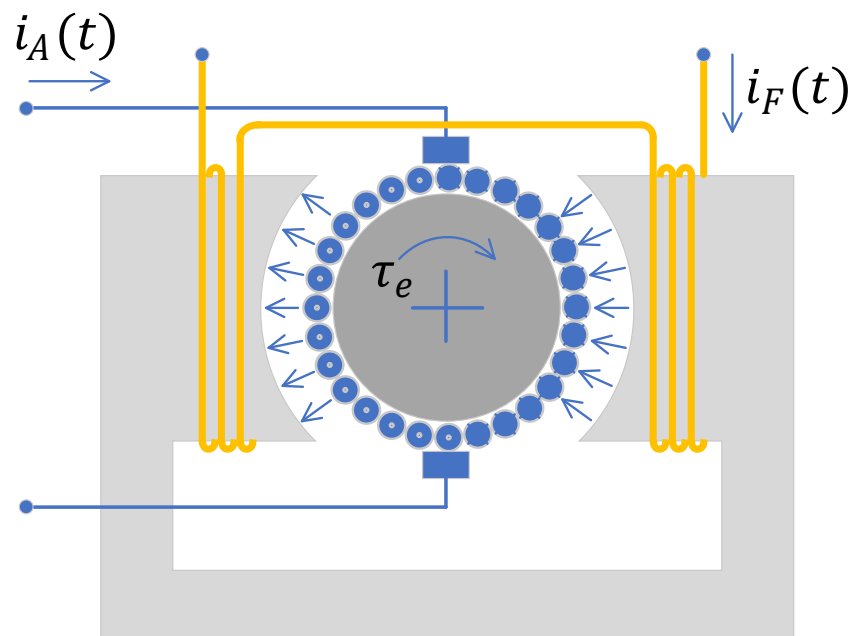


Figure Armature of a dc motor.



Primer 3 - Motor jednosmerne struje



- Prikazan je jednosmerni motor sa nezavisnom pobudom, upravlján strujom rotora.
- Odrediti zavisnost brzine rotora od napona rotora i promene opterećenja motora.
- Pretpostaviti da se radi o opsegu brzina do nominalne ($\omega \leq \omega_n$) i da je fluks u mašini $\Phi_F = const$ (pobudna struja $i_F = I_F = const$).

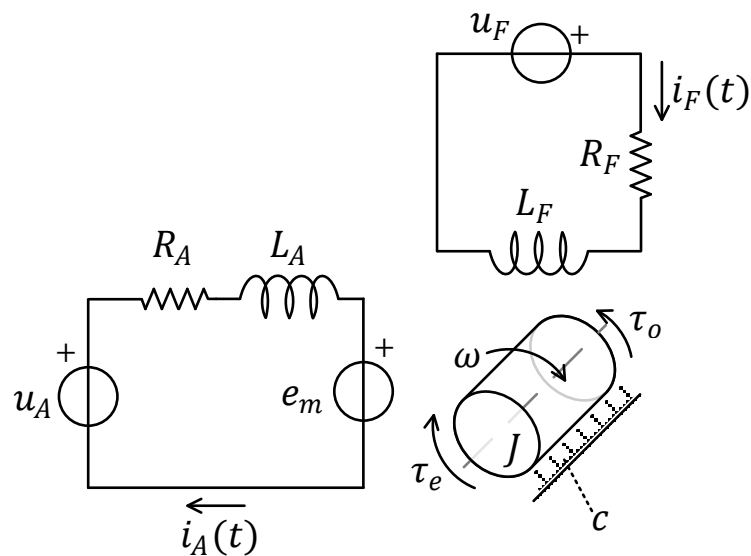
Primer 3 – nastavak ...

$$B = \frac{\Phi(i_F(t))}{A}$$

$$\tau_e(t) = f_e a = (l \cdot B \cdot i_A(t)) a = \left(l \cdot \frac{\Phi(i_F(t))}{A} \cdot i_A(t) \right) a = \gamma \cdot \Phi(i_F(t)) \cdot i_A(t)$$

$$\gamma = \frac{la}{A}$$

$$e_m(t) = lvB = l \cdot \omega(t) a \cdot \frac{\Phi(i_F(t))}{A} = \gamma \cdot \Phi(i_F(t)) \cdot \omega(t)$$



Ako je: $u_F(t) = U_F = \text{const}$, $I_F = \frac{U_F}{R_F}$, $\alpha = \gamma \Phi(I_F)$

$$\tau_e(t) = \alpha \cdot i_A(t)$$

$$e_m(t) = \alpha \cdot \omega(t)$$

$$\frac{di_A(t)}{dt} = \frac{1}{L_A} [-R_A i_A(t) - \alpha \omega(t) + u_A(t)]$$

$$\frac{d\omega(t)}{dt} = \frac{1}{J} [\alpha i_A(t) - c \omega(t) - \tau_o(t)]$$

Primer 3 – nastavak 2

prepisano ...

$$\frac{di_A(t)}{dt} = \frac{1}{L_A} [-R_A i_A(t) - \alpha \omega(t) + u_A(t)]$$

$$\frac{d\omega(t)}{dt} = \frac{1}{J} [\alpha i_A(t) - c \omega(t) - \tau_o(t)]$$

U matričnom zapisu:

$$\begin{bmatrix} \frac{di_A(t)}{dt} \\ \frac{d\omega(t)}{dt} \end{bmatrix} = \begin{bmatrix} \frac{-R_A}{L_A} & \frac{-\alpha}{L_A} \\ \frac{\alpha}{J} & \frac{-c}{J} \end{bmatrix} \cdot \begin{bmatrix} i_A(t) \\ \omega(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L_A} & 0 \\ 0 & -\frac{1}{J} \end{bmatrix} \cdot \begin{bmatrix} u_A(t) \\ \tau_o(t) \end{bmatrix}$$

Model 2. reda == 2 prom. stanja;
2 ulaza

Ako se zamenari induktivnost namotaja $L \ll 1$

$$\frac{d\omega(t)}{dt} = -\left(\frac{\alpha^2}{JR_A} + \frac{c}{J}\right) \omega(t) + \frac{\alpha}{JR_A} u_A(t) - \frac{1}{J} \tau_o(t)$$

Model 1. reda == 1 prom. stanja;
2 ulaza